

Akshit Gangwar

Software Engineer | Backend Developer | Full-Stack Developer

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Summary

Backend-focused software engineer experienced in building and deploying production-grade full-stack applications (50K+ users). Designed API-driven, microservices-style, and serverless systems, with a focus on system architecture and performance-oriented backend engineering.

Work Experience

IT Head, Falak'25 Cultural-Sports Fest, MIT Bengaluru

Aug 2025 – Oct 2025

- Architected and deployed a full-stack fest platform (Next.js, NextAuth, Supabase) with a PostgreSQL-backed backend, achieving 99.9% uptime during peak traffic.
- Achieved 50,000+ visitors in the fest month, 6,500+ registered users, 5,000+ secure ticket sales, and 2,000+ team registrations.
- Built a role-based admin dashboard for 4 types of admins; optimized caching, cutting video load time by 40%.
- Led a 5-member UI/UX team, coordinated cross-team, managing GitHub workflows and delivered within first 2 weeks.
- **Tech Stack:** Next.js, NextAuth, TypeScript, Supabase, Firebase, TailwindCSS, Shadcn, JavaScript, Vercel.
GitHub: github.com/Aksh002/mitblr_falak, Live: falak.mitblr.org.

Education

Manipal Institute of Technology, Bengaluru

2023 – 2027

B.Tech in Computer Science and Engineering — Minor in Data Science — CGPA: 8.78/10

Technical Skills

- **Languages:** C++, C, TypeScript, JavaScript, Python
- **Frontend:** Next.js, React.js, TailwindCSS, Recoil, Motion.dev
- **Backend/Auth:** Node.js, Express.js, FastAPI, REST APIs, WebSockets, Webhooks, JWT, OAuth, NextAuth/Auth.js
- **Database and ORM:** PostgreSQL, MongoDB, MySQL, SQLite, Prisma
- **Cloud/DevOps:** Docker, AWS, Cloudflare Workers, Vercel, CI/CD, Git, GitHub, Postman
- **Core CS:** Data Structures and Algorithms, OOP, DBMS, System Design, and Computer Networks

Projects

Learning-in-Public Serverless Blogging Platform

Feb 2025 – Mar 2025

- Architected a **Cloudflare Workers edge** blogging platform evolving beyond Medium-style publishing into a learning-in-public knowledge system with Markdown posts, drafts, series, revisions, highlights, and reader notes.
- Designed **Stateless REST APIs** with JWT authentication at the edge, enabling distributed request handling through Hono.js while reducing infrastructure overhead and latency (by 40%). Live: live-blogging.vercel.app.
- Integrated Prisma with PostgreSQL under serverless constraints, using connection pooling, supporting scalable user, post, tag, comment, follow, like, bookmark, and publishing workflows. GitHub: github.com/Aksh002/learning-in-public-blogger.
- **Tech Stack:** Hono.js, Cloudflare Workers, Prisma, JWT, PostgreSQL, React, TypeScript, Recoil, Vite, Tailwind CSS.

Payments Infrastructure Platform

Jul 2025 – Sep 2025

- Designed a modular monorepo architecture (Turborepo) simulating microservice boundaries for payments, authentication, and transaction processing, supporting secure P2P and P2M transactions across user and merchant portals.
- Implemented multi-provider authentication via NextAuth.js, designed asynchronous payment flows using Express.js webhooks and job queues, and deployed Dockerized services with CI/CD pipelines for scalability and minimal downtime.
- **Tech Stack:** Next.js, Node.js, Express.js, TypeScript, PostgreSQL (Prisma), NextAuth.js, Tailwind CSS, Motion.dev, Webhooks, Docker, CI/CD, AWS, Turborepo. GitHub: github.com/Aksh002/Paytm-project-adv.

AI-Powered Manim Video Generation Platform

Feb 2026 – Jun 2026

- Built a full-stack prompt-to-video platform that converts educational prompts into validated Manim code with MP4 preview, download, signed artifact delivery, and S3/R2-compatible storage. Live: manim-ai-alpha.vercel.app.
- Designed FastAPI services for generation, rendering, status tracking, regeneration, cancellation, worker health, cleanup, and artifact access using Redis, RQ, structured job metadata, and Prisma/PostgreSQL-backed ownership.
- Implemented AST validation, deterministic repair, LLM-based auto-fix retries, quality evaluation, and sandboxed Manim execution with Docker/service-based renderer isolation. GitHub: github.com/Aksh002/Manim-ai.
- Delivered a Next.js + Monaco editor workflow with Auth.js login, usage-based free credits, encrypted BYOK OpenAI-compatible LLM keys, render history, repair timeline, Vercel-ready deployment, and Railway backend.
- **Tech Stack:** FastAPI, Python, Next.js, TypeScript, Manim, Auth.js, Prisma, PostgreSQL, Redis, RQ, Docker, S3/R2.

Relevant Coursework

- Web Development and DevOps Cohort by Harkirat (2024) — github.com/Aksh002/notes-100xdevs
- Introduction to Java and Object-Oriented Programming — coursera.org/account/accomplishments/verify/YUC1Q2700IW8